Tutorial 2 : MIT 6.041 Probability Systems Analysis

1. (a) 13th card → 4 choices out of 52 → take 1

12th card → 51 choices (3 remaining kings included) out of 51

11th card → 50 choices out of 50

...

P(13th card is a king) = P(13th card is a king ∩ other cards can be anything) = $\frac{4}{52} \cdot \frac{51}{51} \cdot \frac{50}{50} \cdots \frac{40}{40} = \frac{1}{13}$

(b) 13th card is first king → 4 choices out of 52

12th card is not a king → 48 choices out of 51

11th card is not a king → 47 choices out of 50

...

P(13th card is 1st king) = P(13th card is a king ∩ no other card is a king) = $\frac{4 \cdot 48 \cdot 47 \cdots 37}{52 \cdot 51 \cdot 50 \cdots 40}$

2. $p_X(x) = \frac{x^2}{a}$ for $x \in \{-3, -2, -1, 1, 2, 3\}$

and $P_X(x) = 0 \; \forall \; x \notin \{-3, -2, -1, 1, 2, 3\}$

where $a > 0$

(a) $\sum_x p_x(x) = 1 \Rightarrow \frac{1}{a}\sum_x x^2 = 1$

$\Rightarrow \frac{1}{a}(18+8+2) = 1 \Rightarrow a = 28$

(b) $z = x^2 \Rightarrow z \in \{1, 4, 9\}$

$z = 1 \Rightarrow x \in \{-1, 1\}$

$\Rightarrow p_z(z=1) = p_x(x \in \{-1,1\}) = 1/14$

$z = 4 \Rightarrow x \in \{-2, 2\}$

$\Rightarrow p_z(z=4) = p_x(x \in \{-2,2\}) = 2/7$

$z = 9 \Rightarrow x \in \{-3, 3\}$

$\Rightarrow p_z(z=9) = p_x(x \in \{-3,3\}) = 9/14$

3. # ways to split 90 students into 3 groups of 30

$$= \frac{90!}{30!\,30!\,30!}$$

Joe and Jane end up in same class:

Pick a class (3 choices) → Divide remaining 88 people into 3 groups $= 3 \times \frac{88!}{28!\,30!\,30!}$

$$\Rightarrow \text{probability} = \frac{3 \times 88!}{28!\,30!\,30!} \times \frac{30!\,30!\,30!}{90!} = \frac{29}{89}$$

4. # to choose 7 cards out of 52 = $^{52}C_7$

ways to draw 7 cards with 3 aces

= Select 3 aces out of 4 ($^4C_3$) → select remaining 4 cards out of remaining 48 non aces ($^{48}C_4$)

⇒ Probability = $\frac{^4C_3 \cdot {}^{48}C_4}{^{52}C_7}$